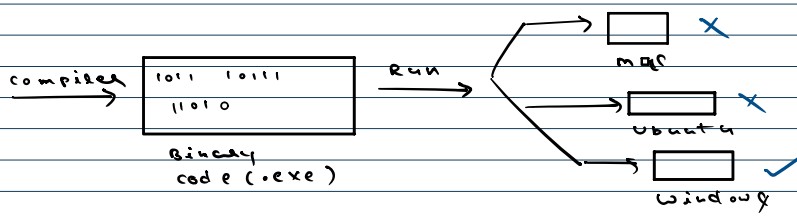


C++ → It is not platform independent

```
#include <iostream>
using namespace std;
int main() {
    int a = 9;
    int b = 10;
    int c = a + b;
    cout << c;
}
```

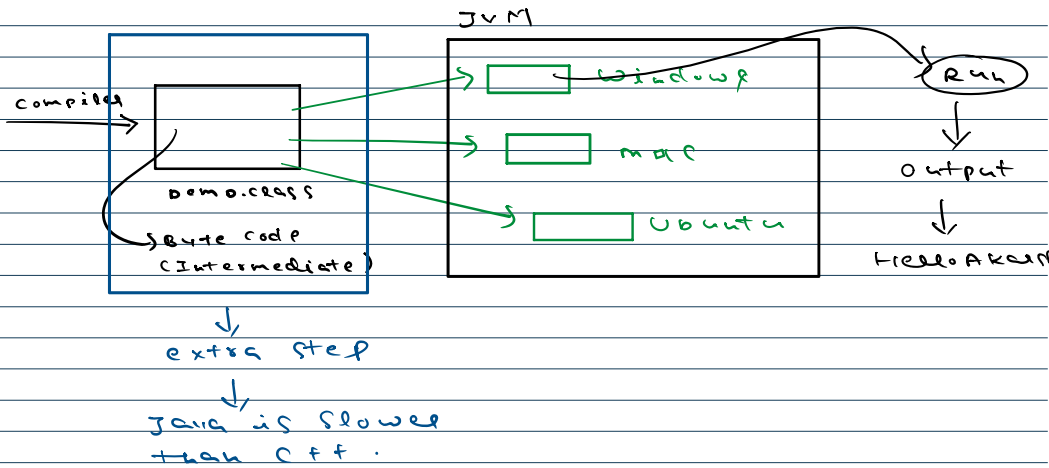
→ Windows



source code

Java → It is platform independent

```
public class Demo {
    public static void main(String args[]) {
        System.out.println("Hello Akarsh");
    }
}
```

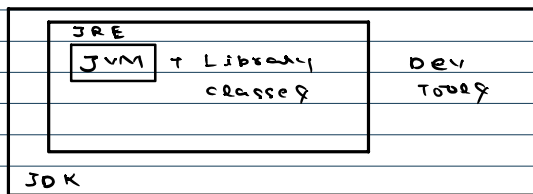


source code → .java
Byte code → .class

extra step

↓
Java is slower than C++.

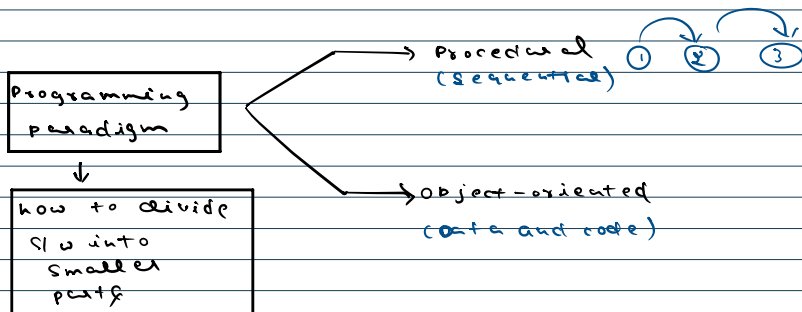
- JVM → Java Virtual Machine
- JRE → Java Runtime Environment
- JDK → Java Development Kit



Till now → Function oriented i.e. as same as C.

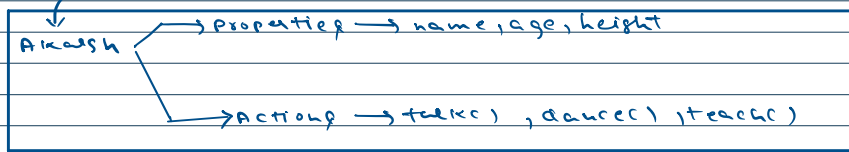
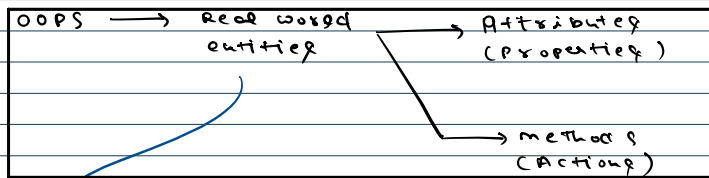
Object-Oriented Programming

→ clean, structured and reusable code.



OOPS → Real world entities → Attributes (properties)





objects → ?
 ↓
 When we combine data and functions, that operate on this data, they form object.

Akash

class → It is the blueprint of the object.
 Object → It is an instance of a class.

Human

```
public class Student {
    2 usages
    String name;
    2 usages
    int age;
}
```

```
public class StudentClient {
    public static void main(String args[]) {
        Student s = new Student();
        s.name = "Akash";
        s.age = 25;

        Student s1 = new Student();
        s1.name = "Abhi";
        s1.age = 27;
    }
}
```

